Astrodynamics 2

Homework #04

2020.11.19

김나은

Homework 4

1. Find three position vectors (R1, R2, R3) using Gauss' method.
2. Determine the velocity $\mathbf{V}_2$ associated with $\mathbf{R}_2$.
3. Use the Algorithm 9 to determine the orbital elements using ($\mathbf{R}_2$, $\mathbf{V}_2$).

Example 7-2. Determining an Initial Orbit Using Angles Only.

GIVEN: Three sets of right ascension-declination observations (topocentric) from a station at 40°N, 110°W, alt = 2000 m, on August 20, 2007. (Use observations number 6, 8, and 14.)

Obs. #	h min s	α_t	δ_t
1	11: 15: 00.00	–63.2154821	–28.0157802
2	11: 20: 00.00	–54.1249012	–22.6156514
3	11: 25: 00.00	–44.3019142	–15.8428032
4	11: 30: 00.00	–33.0588410	–7.2056382
5	11: 35: 00.00	–19.1381645	3.9172775
6	11: 40: 00.00	–0.4172870	17.4626616
7	11: 45: 00.00	25.2404412	30.2936123
8	11: 50: 00.00	55.0931551	36.5731946
9	11: 55: 00.00	80.6761692	35.5325741
10	12: 00: 00.00	98.7739537	31.1314513
11	12: 05: 00.00	111.4115235	25.9983390
12	12: 10: 00.00	120.7634587	21.0068603
13	12: 15: 00.00	128.1416909	16.3518151
14	12: 20: 00.00	134.2826693	12.0351097
15	12: 25: 00.00	139.6133831	8.0147326

FIND: Position and velocity vectors for the observation using Gauss's method.

Homework 4

1. Find three position vectors (R1, R2, R3) using Gauss' method.

- Observation station
  - 40 ºN, 110 ºW, alt=2000 m

- Observation data
  - August 20, 2007

Obs. #	h min s	α_t	δ_t
6	11:40:00.00	-0.4172870	17.4626616
8	11:50:00.00	55.0931551	36.5731946
14	12:20:00.00	134.2826693	12.0351097

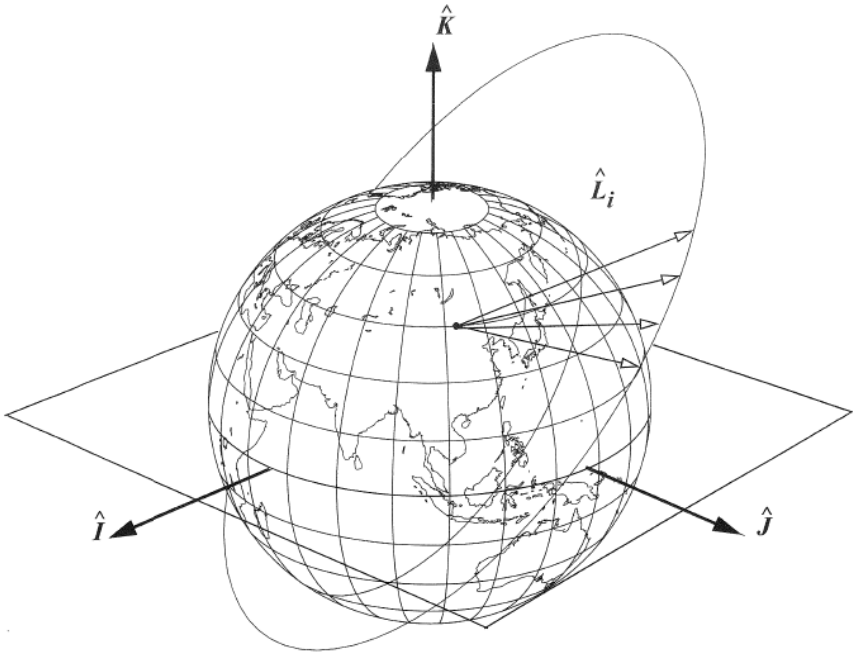


Figure. Angles-only observations
(Vallado, 2013)

Homework 4

- Gauss method
  - Assume 3 position vectors lie in a single plane

$$c_1\vec{r}_1 + c_2\vec{r}_2 + c_3\vec{r}_3 = 0$$

  - f & g function

$$\vec{r}_i = f_i\vec{r}_2 + g_i\vec{v}_2 \quad (i = 1,3)$$

  - Find c

$$c_1 = \frac{g_3}{f_1 g_3 - f_3 g_1}, \qquad c_3 = \frac{-g_1}{f_1 g_3 - f_3 g_1}$$

  - f & g function with respect to time τ with t_2 $(epoch)$

$$f_i = 1 - \frac{1}{2}u\tau_i^2 + \theta(\tau_i^3)$$

$$g_i = \tau_i - \frac{1}{6}u\tau_i^3 + \theta(\tau_i^4)$$

$$\left(where\ \tau_i = t_i - t_2\ (i = 1,3), \qquad u = \frac{\mu}{r_2^3}\right)$$

Homework 4

- Gauss method
  - Find c with respect to time τ with t_2 $(epoch)$

$$c_1 = a_1 + a_{1\mathrm{u}}\mathrm{u}, \qquad c_3 = a_3 + a_{3u}u$$

$$\left(where \quad a_1 = \frac{\tau_3}{\tau_3 - \tau_1}, \qquad a_{1\mathrm{u}} = \frac{\tau_3[(\tau_3 - \tau_1)^2 - \tau_3^2]}{6(\tau_3 - \tau_1)} \right.$$
$$\left. a_3 = -\frac{\tau_1}{\tau_3 - \tau_1}, \qquad a_{3\mathrm{u}} = -\frac{\tau_1[(\tau_3 - \tau_1)^2 - \tau_1^2]}{6(\tau_3 - \tau_1)} \right)$$

  - $\vec{r} = \rho\hat{L} + \vec{r}_{site}$

$$\vec{L}\begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = [\vec{r}_{site,1} \quad \vec{r}_{site,2} \quad \vec{r}_{site,3}]\begin{bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{bmatrix}$$

  - Find ρ_2

$$\rho_2 = d_1 + d_2 u$$
$$(where\ d_1 = M_{21}a_1 - M_{22} + M_{23}\ a_3, d_2 = M_{21}a_{1u} + M_{23}a_{3u})$$

  - Find r_2

$$r_2^8 - \left(d_1^2 + 2Cd_1 + r_{site,2}^2\right)r_2^6 - 2\mu(Cd_2 + d_1d_2)r_2^3 - \mu^2 d_2^2 = 0$$

Homework 4

- Algorithm 51
  - Compute τ

$$\tau_1 = JD_1 - JD_2, \tau_3 = JD_3 - JD_2$$

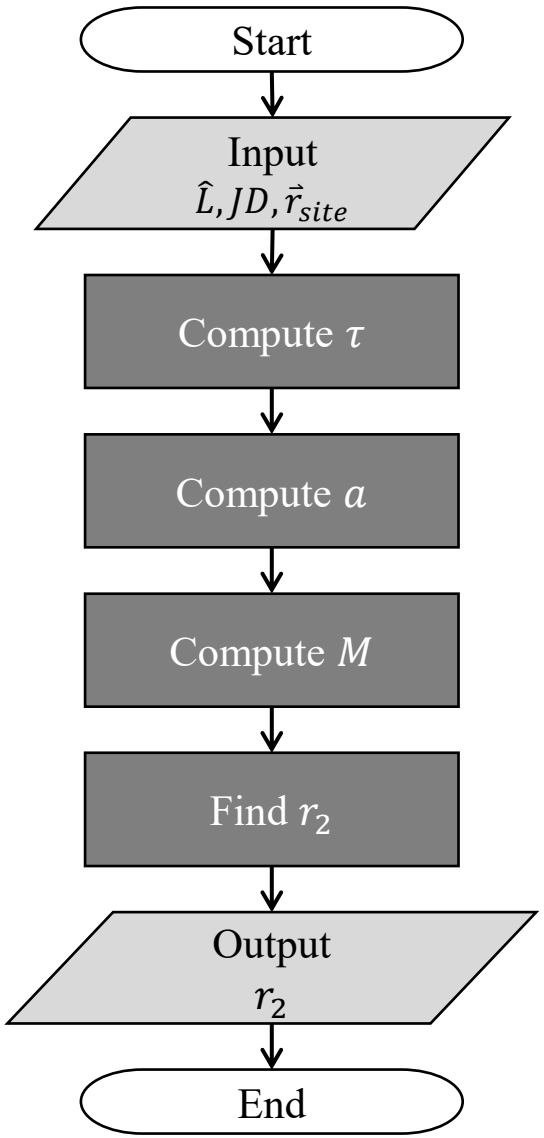
  - Compute a

$$a_1 = \frac{\tau_3}{\tau_3 - \tau_1},\ a_{1\mathrm{u}} = \frac{\tau_3\left[(\tau_3-\tau_1)^2 - \tau_3^2\right]}{6(\tau_3-\tau_1)}$$

$$a_3 = -\frac{\tau_1}{\tau_3 - \tau_1},\ a_{3\mathrm{u}} = -\frac{\tau_1\left[(\tau_3-\tau_1)^2 - \tau_1^2\right]}{6(\tau_3-\tau_1)}$$

  - Compute L (Line of site)

$$\widehat{L_i} = \begin{bmatrix} \cos(\delta_i)\cos(\alpha_i) \\ \cos(\delta_i)sin(\alpha_i) \\ \sin(\delta_i) \end{bmatrix}$$



Homework 4

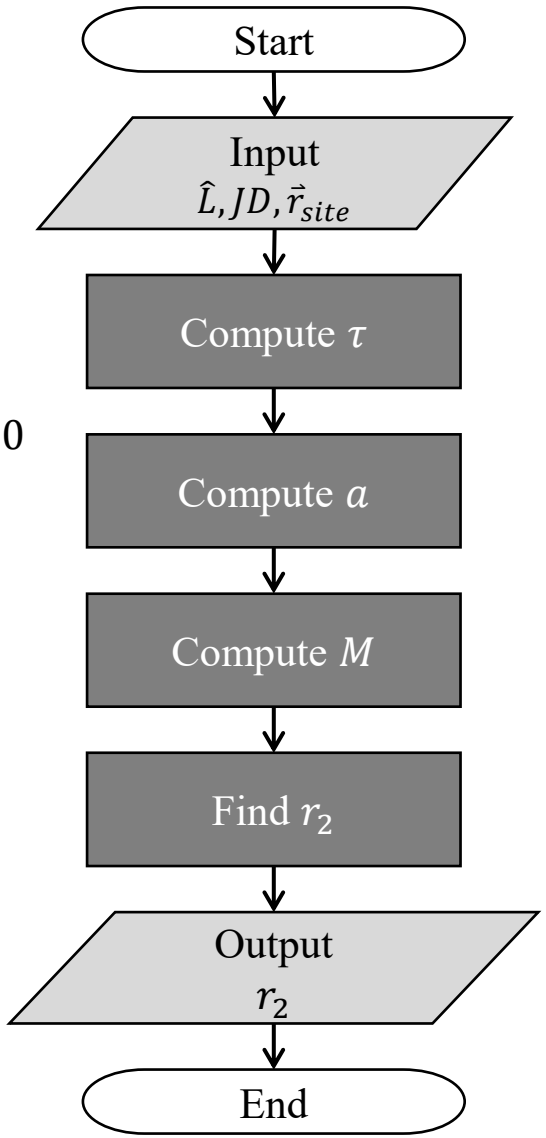
- Algorithm 51
  - Compute M

$$[M] = [L^{-1}][r_{site}]$$

  - Find r_2

$$r_2^8 - \left(d_1^2 + 2Cd_1 + r_{site,2}^2\right)r_2^6 - 2\mu(Cd_2 + d_1d_2)r_2^3 - \mu^2 d_2^2 = 0$$

$$(where\ d_1 = M_{21}a_1 - M_{22} + M_{23}\ a_3, d_2 = M_{21}a_{1u} + M_{23}a_{3u}$$
$$C = \hat{L}_2\vec{r}_{site,2})$$



Homework 4

- Matlab code
  - Input

```
a6= -0.4172870; d6=17.4626616;
a8=55.0931551; d8=36.5731946;
a14=134.2826693; d14=12.0351097;
```

  - Compute τ & a

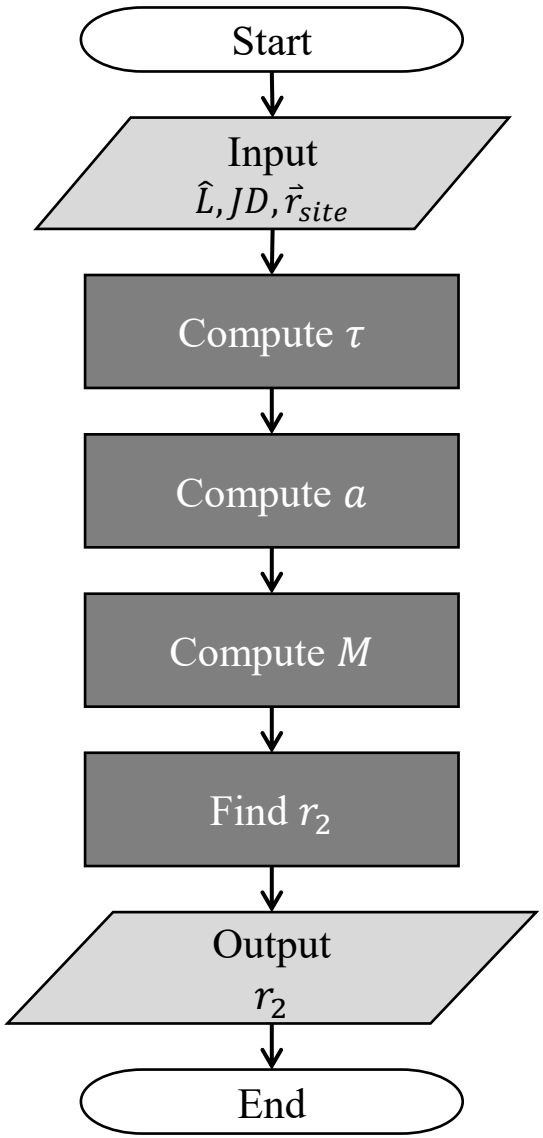
```
t1=-600;
t3=1800;
a1=t3/(t3-t1); a1u=t3*((t3-t1)^2-t3^2)/(6*(t3-t1));
a3=-t1/(t3-t1); a3u=-t1*((t3-t1)^2-t1^2)/(6*(t3-t1));
```

  - Compute L (Line of site)

```
L6=[cos(d(d6))*cos(d(a6)) cos(d(d6))*sin(d(a6)) sin(d(d6))].';
L8=[cos(d(d8))*cos(d(a8)) cos(d(d8))*sin(d(a8)) sin(d(d8))].';
L14=[cos(d(d14))*cos(d(a14)) cos(d(d14))*sin(d(a14)) sin(d(d14))].';
L=[L6 L8 L14];
```

L=

0.9539	0.4596	-0.6829
-0.0069	0.6586	0.7002
0.3001	0.5958	0.2085



Homework 4

- Algorithm 51
  - ➢ Compute M

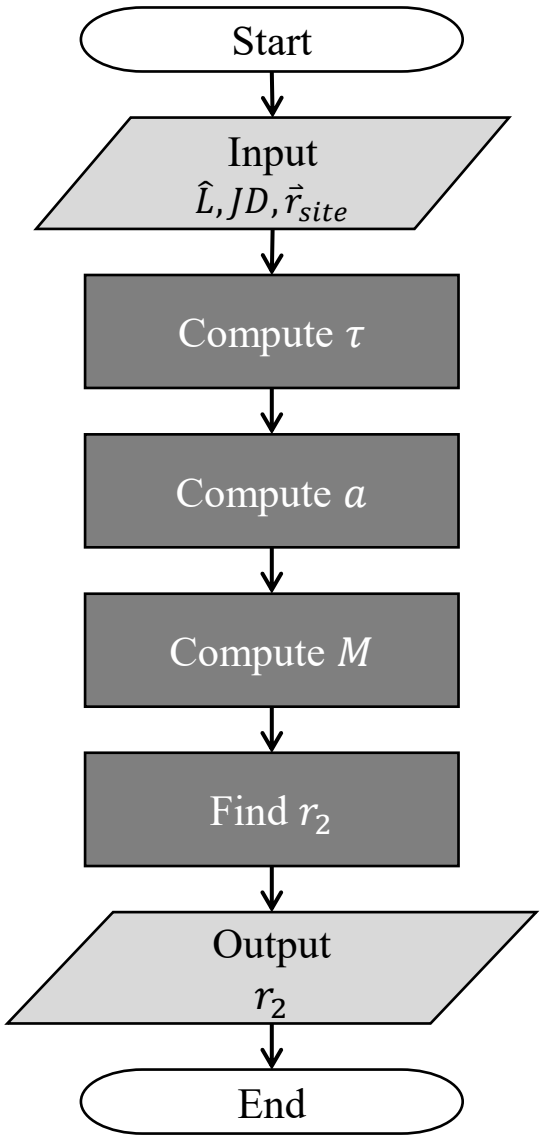
Find $r_{site_{ECEF}}$

$$C = \frac{R}{\sqrt{1-e^2 \sin^2(\phi)}} \quad S = C * (1 - e^2)$$

$$r_\delta = (C + h) * cos\phi, \; r_k = (S + h) * sin\phi$$

$$r_{site_{ECEF}} = \begin{bmatrix} r_\delta \cos(\theta) \\ r_\delta \sin(\theta) \\ r_k \end{bmatrix}$$

```
R=6371; e=0.0167; al=2; gd=40; long=250;
C=R/sqrt(1-e^2*sin(d(gd))); S=C*(1-e^2);
r_del=(C+al)*cos(d(gd)); r_k=(S+al)*sin(d(gd));
r_site=[r_del*cos(d(long)) r_del*sin(d(long)) r_k].';
```



Homework 4

- Algorithm 51
  - Compute M

Find $r_{site_{ECI}}$

```
dcm6 = dcmeci2ecef('IAU-2000/2006',[2007 8 20 11 40 00]);
dcm8 = dcmeci2ecef('IAU-2000/2006',[2007 8 20 11 50 00]);
dcm14 = dcmeci2ecef('IAU-2000/2006',[2007 8 20 12 20 00]);
r_site_ECI=[dcm6'*r_site dcm8'*r_site dcm14'*r_site];
```

$r_{site_{ECI}} =$

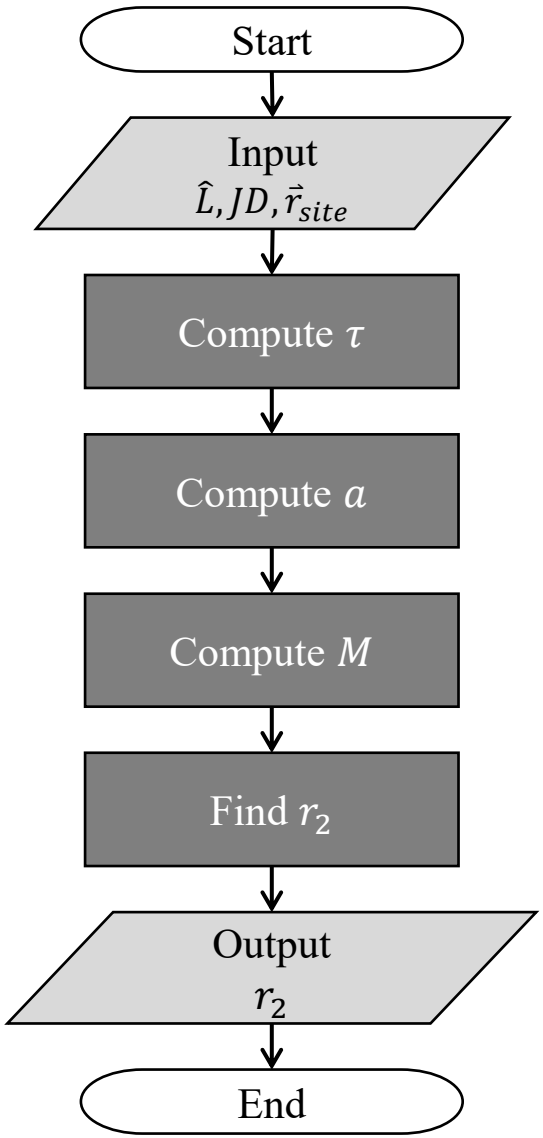
4.0823e+03	3.9610e+03	3.5528e+03
2.6832e+03	2.8590e+03	3.3524e+03
4.0925e+03	4.0926e+03	4.0929e+03

Find M

```
M=L*r_site_ECI;
```

$M =$

-2.0845e+04	-1.9142e+04	-1.4962e+04
2.3996e+04	2.2579e+04	1.9052e+04
-1.8947e+04	-1.7345e+04	-1.3281e+04

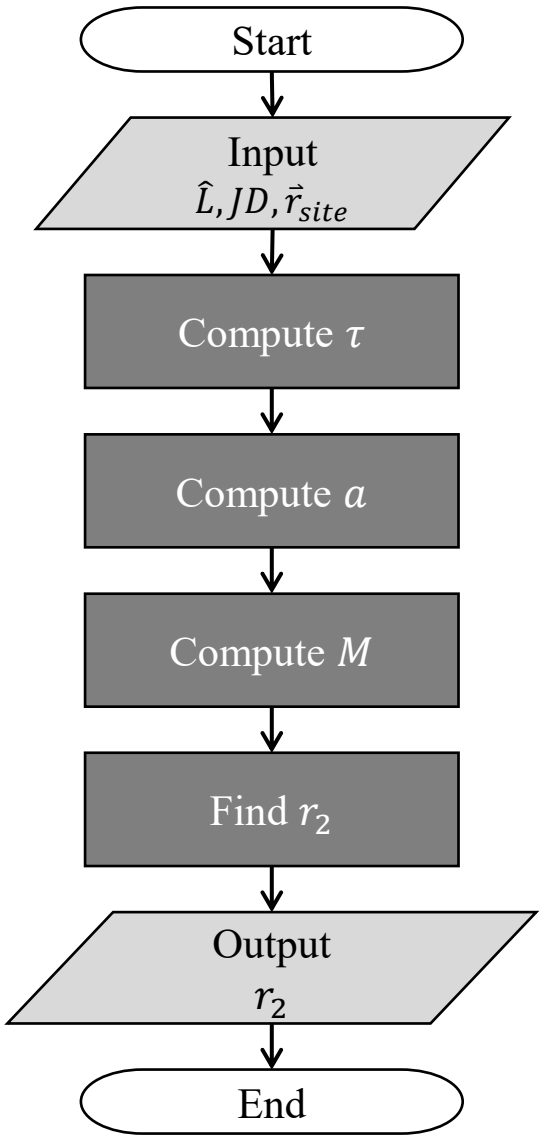


Homework 4

- Algorithm 51
  - Find r_2

```
d1=M(2,1)*a1-M(2,2)+M(2,3)*a3;
d2=M(2,1)*a1u+M(2,3)*a3u;
C=L(:,2).'*r_site_ECI(:,2);
r_site2=norm(r_site_ECI(:,2));
mu=398600.4418;
p=[1, 0,-(d1^2+2*C*d1+r_site2^2) 0 0 -2*mu*(C*d2+d1*d2) 0 0 -mu^2*d2^2];
r2=roots(p);
%Using first positive root
r2=r2(4);
```

r_2 =10518.73004350932



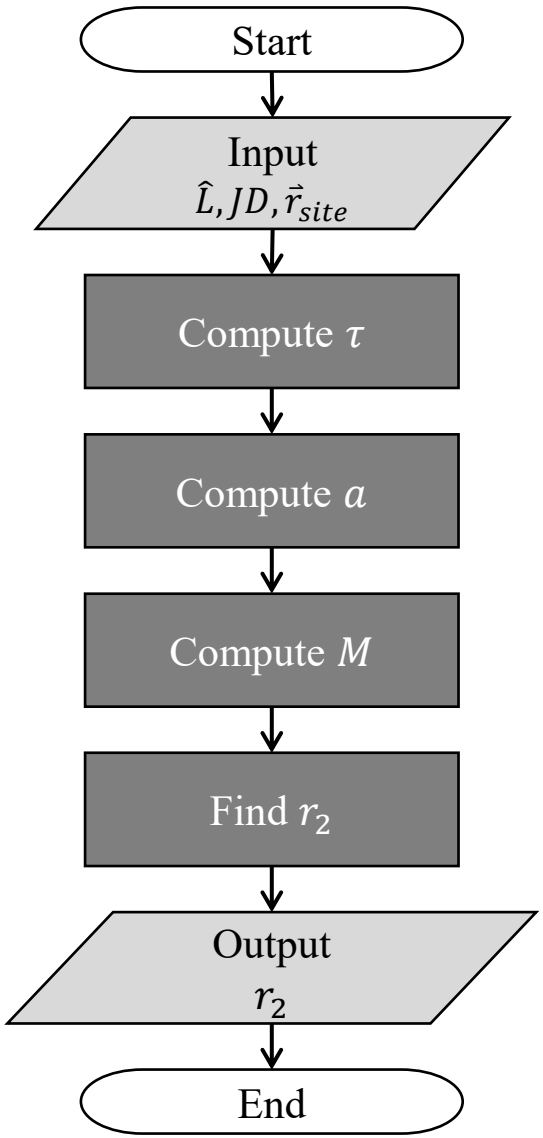
Homework 4

- Algorithm 51
  - Fine $\vec{r}$

$$\vec{L}\begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = [\vec{r}_{site,1} \quad \vec{r}_{site,2} \quad \vec{r}_{site,3}]\begin{bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{bmatrix}$$

```
u=mu/r2^3;
c=[a1+a1u*u -1 a3+a3u*u].';
ro=M*-c./c;
n=1;
r=[];
for n=1:3
    r_n=[ro(n) ro(n) ro(n)].'.*L(:,n)+r_site_ECI(:,n);
    r(n,:)=[r_n];
end
```

$\vec{r}_1=$	8.1224e+03	2.6537e+03	5.3635e+03
$\vec{r}_2=$	5.9089e+03	5.6505e+03	6.6181e+03
$\vec{r}_3=$	-3.2385e+03	1.0316e+04	6.1666e+03



Homework 4

2. Determine the velocity $\mathbf{V}_2$ associated with $\mathbf{R}_2$.

- Gibbs algorithm

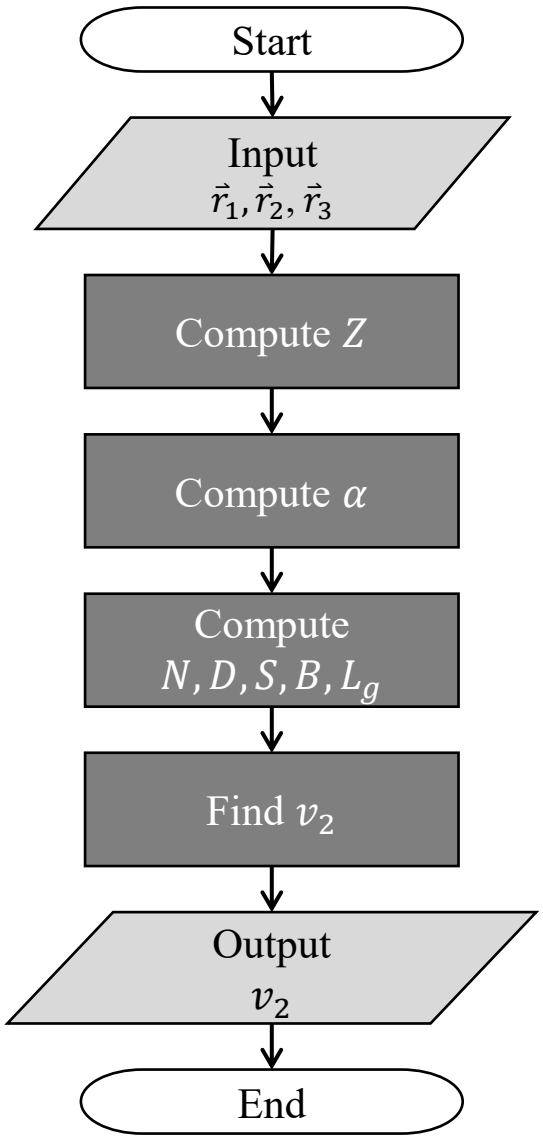
- Compute Z

$$\vec{Z}_{12} = \vec{r}_1 \times \vec{r}_2 \qquad \vec{Z}_{23} = \vec{r}_2 \times \vec{r}_3 \qquad \vec{Z}_{31} = \vec{r}_3 \times \vec{r}_1$$

- Compute α

$$\alpha_{cop} = \mathrm{SIN}^{-1}\left(\frac{\vec{Z}_{23} \cdot \vec{r}_1}{|\vec{Z}_{23}||\vec{r}_1|}\right)$$

$$\cos(\alpha_{12}) = \frac{\vec{r}_1 \cdot \vec{r}_2}{|\vec{r}_1||\vec{r}_2|} \qquad \cos(\alpha_{23}) = \frac{\vec{r}_2 \cdot \vec{r}_3}{|\vec{r}_2||\vec{r}_3|}$$



Homework 4

2. Determine the velocity $\mathbf{V}_2$ associated with $\mathbf{R}_2$.

- Compute N, D, S, B, L_g

$$\vec{N} = r_1\vec{Z}_{23} + r_2\vec{Z}_{31} + r_3\vec{Z}_{12}$$

$$\vec{D} = \vec{Z}_{12} + \vec{Z}_{23} + \vec{Z}_{31}$$

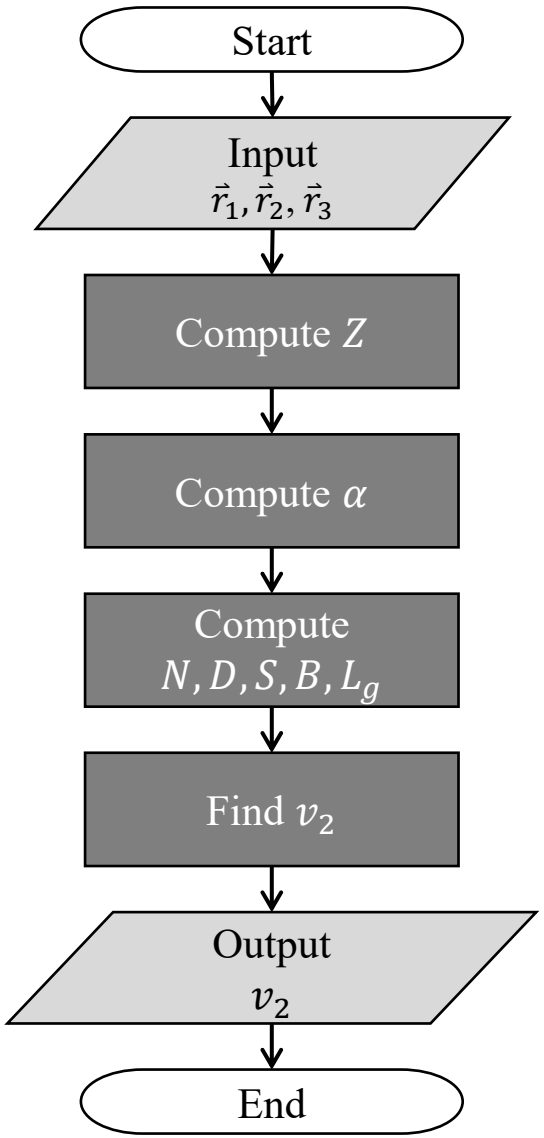
$$\vec{S} = (r_2 - r_3)\vec{r}_1 + (r_3 - r_1)\vec{r}_2 + (r_1 - r_2)\vec{r}_3$$

$$\vec{B} = \vec{D} \times \vec{r}_2$$

$$L_g = \sqrt{\frac{\mu}{ND}}$$

- Find v_2

$$\vec{v}_2 = \frac{L_g}{r_2}\vec{B} + L_g\vec{S}$$



Homework 4

- Matlab code
  - Compute Z

```
z12=cross(r(1,:),r(2,:));
z23=cross(r(2,:),r(3,:));
z31=cross(r(3,:),r(1,:));
```

  - Compute N, D, S, B, L_g

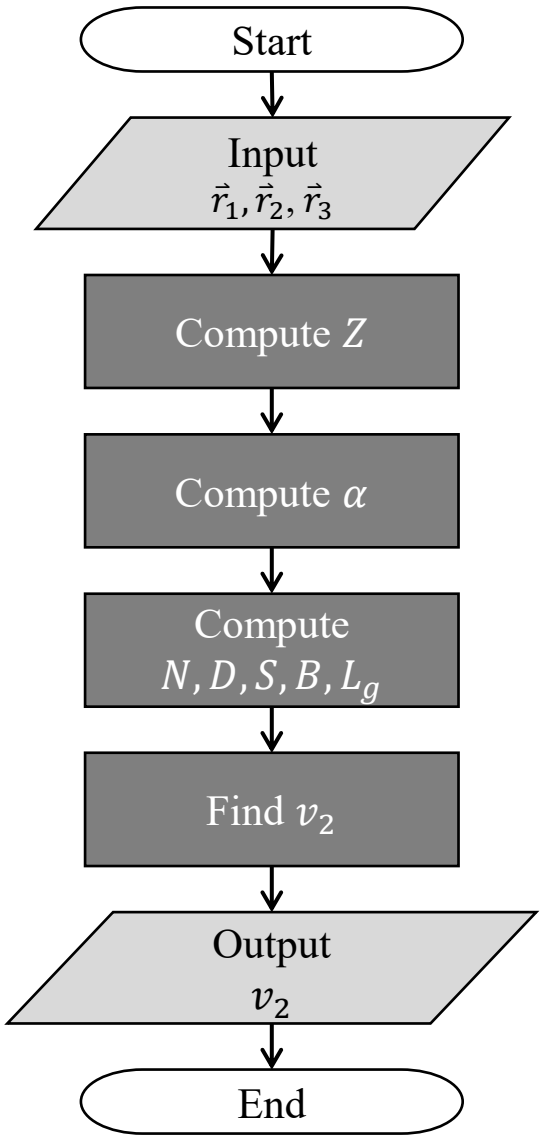
```
r1=norm(r(1,:)); r2=norm(r(2,:)); r3=norm(r(3,:));
N=r1*z23+r2*z31+r3*z12;
D=z12+z23+z31;
S=(r2-r3)*r(1,:)+(r3-r1)*r(2,:)+(r1-r2)*r(3,:);
B=cross(D,r(2,:));
Lg=sqrt(mu/(norm(N)*norm(D)));
```

  - Find v_2

```
v2=Lg/r2.*B+Lg.*S;
```

$v_2 =$

-4.4899	4.6300	1.4870



Homework 4

3. Use the Algorithm 9 to determine the orbital elements using ($\mathbf{R}_2$, $\mathbf{V}_2$).

➢ Algorithm 9 ***RV2COE*** ($\vec{r}_{IJK}$, $\vec{v}_{IJK} \Rightarrow p, a, e, i, \Omega, \omega, \nu(u, \lambda_{true}, \tilde{\omega}_{true})$)

$$\vec{h} = \vec{r} \times \vec{v} \qquad h = |\vec{h}|$$

$$\vec{n} = \hat{K} \times \vec{h}$$

$$\vec{e} = \frac{\left(v^2 - \frac{\mu}{r}\right)\vec{r} - (\vec{r} \cdot \vec{v})\vec{v}}{\mu} \qquad e = |\vec{e}|$$

$$\xi = \frac{v^2}{2} - \frac{\mu}{r}$$

IF $e \neq 1.0$ THEN

$$a = -\frac{\mu}{2\xi}$$

$$p = a(1 - e^2)$$

ELSE

$$p = \frac{h^2}{\mu} \text{ and } a = \infty$$

$$\cos(i) = \frac{h_K}{|\vec{h}|}$$

$$\cos(\Omega) = \frac{n_I}{|\vec{n}|} \qquad \text{IF}(n_J < 0)\text{THEN } \Omega = 360^\circ - \Omega$$

$$\cos(\omega) = \frac{\vec{n} \cdot \vec{e}}{|\vec{n}||\vec{e}|} \qquad \text{IF}(e_K < 0)\text{THEN } \omega = 360^\circ - \omega$$

$$\cos(\nu) = \frac{\vec{e} \cdot \vec{r}}{|\vec{e}||\vec{r}|} \qquad \text{IF}(\vec{r} \cdot \vec{v} < 0)\text{THEN } \nu = 360^\circ - \nu$$

Special Cases

Elliptical equatorial:

$$\cos(\tilde{\omega}_{true}) = \frac{e_I}{|\vec{e}|} \qquad \text{IF}(e_J < 0)\text{THEN } \tilde{\omega}_{true} = 360^\circ - \tilde{\omega}_{true}$$

Circular Inclined:

$$\cos(u) = \frac{\vec{n} \cdot \vec{r}}{|\vec{n}||\vec{r}|} \qquad \text{IF}(r_K < 0)\text{THEN } u = 360^\circ - u$$

Circular equatorial:

$$\cos(\lambda_{true}) = \frac{r_I}{|\vec{r}|} \qquad \text{IF}(r_J < 0)\text{THEN } \lambda_{true} = 360^\circ - \lambda_{true}$$

Homework 4

- Matlab code

```
E=norm(v2)^2/2-mu/r2;
h=cross(r(2,:),v2);
a=-mu/(2*E);
e=((norm(v2)^2-mu/r2).*r(2,:)-(r(2,:)*v2.').*v2)./mu;
i=rad2deg(acos(h(3)/norm(h)));
n=cross([0 0 1], h);
om=360-rad2deg(acos(n(1)/norm(n)));
long_p=rad2deg(acos(n*e.'/(norm(n)*norm(e))));
true=rad2deg(acos(e*r(2,:).'/(norm(e)*norm(r(2,:)))));
```

```
a =

   1.2463e+04


e =

    0.2059


i =

   40.1390


om =

  329.9883


long_p =

   28.2475


true =

   49.1759
```

Thank you for listening